

- 40** Some vaccines do not contain antigens. The vaccines contain a molecule of mRNA. Cells in the immune system use the mRNA molecule to make a protein antigen.

The statements describe the stages of how mRNA vaccines work when they enter a cell of the immune system.

- 1 B-lymphocytes and T-lymphocytes with complementary receptors bind to the protein.
- 2 Lymphocytes differentiate into memory cells that give long lasting immunity.
- 3 Ribosomes translate the mRNA molecule to make a protein.
- 4 The cell displays the protein on its cell surface membrane.

What is the correct order of the stages of how mRNA vaccines work?

- A** 1 → 2 → 4 → 3
- B** 3 → 1 → 2 → 4
- C** 3 → 4 → 1 → 2
- D** 4 → 1 → 2 → 3

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